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Moneyball - Finding Low Cost Mission Architectures By Design Space Exploration Using Pattern Languages and Houbolt Questions

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Abstract

With the advent of the Commercial Lunar Payload Services Program and the envisaged increase in human activity in cis-lunar space and on the lunar surface, space mission designers will need to create economically feasible and environmentally sustainable mission architectures for complex-mode missions.

The current standard for space mission design involves experts creating alternative mission architectures from their knowledge base and performing trade-studies to select the most promising candidates. This process searches only a limited part of the design space as it relies on human imagination to generate the concepts and thus it risks missing the most cost effective mission architectures.

This paper demonstrates a model-based method to exhaustively chart the design space for space missions using pattern languages and Houbolt questions. Pattern languages are a way of expressing ConOps as a series of characters arranged together according to a set of simple grammatical rules. Houbolt questions are key choices which are made when designing a space mission such as the propulsion choice, technological development strategy, spacecraft production method and reusability considerations. Pattern languages and Houbolt questions are used for combinatorial exploration of the design space.

Cost estimations and assessments of risk, uncertainty are performed on each mission architecture in the design space using a modified version of TruePlanning cost estimation software. The method returns a set of promising mission architectures which is more likely to contain the solution which is the optimum of the optimum than the existing method of ideation of concepts of operations based on expertise, and can create new unimagined low cost mission architectures.

The method is to be tested on a case study of the European Rosetta mission: A comet orbiter/lander mission which visited asteroid 67p in 2016. The results show that exhaustive design space exploration can find the pareto-optimal set of mission architectures for this case study, and can produce some interesting alternative solutions for space missions.

Keywords: Pattern Language, Houbolt Question, System Design, Cost Engineering, TruePlanning, Mission Architecture, Optimisation, Model Based Engineering

1. Introduction

It will not surprise the learned reader that spaceflight is an extraordinarily expensive activity. However, it is interesting to see different measures of just how expensive it is. For example, the Apollo program retrieved 382kg of samples from the lunar surface[1].

The total cost of the Apollo program was \$257 billion (2020 dollars) [2]. Therefore, if NASA were to pay for the Apollo missions by selling the rocks they brought back, they would need to be sold for \$672 million per kg. For comparison, rough Earth diamonds fetched around \$12 million per kg in 2020 [3]. Which means that even if the moon was made of diamonds, NASA could not recoup their investment this way.

Of course, the Apollo lunar missions were not primarily concerned with maximising sample return, or minimising costs, but this example shows that if we are to create a economically feasible cis-lunar economy involving frequent cargo and human transport missions to and **from** the moon, then Apollo-like mission architectures will be infeasibly expensive. We need to design new, much cheaper lunar mission architectures.

Regarding the life cycle cost of space systems, the decisions with the highest cost-impact occur in the early design phases. Up to 70% of the total costs of a mission are committed before the end of Phase-A [4]. Therefore the interests of this paper are in the preliminary and conceptual design of low-cost space missions in pre-Phase-A and Phase-A.

This paper presents a preliminary design method to find low cost mission architectures using Houbolt Questions: a novel method of modelling space missions to enable combinatorial design space exploration, and use of the Advanced Mission Cost Model to estimate the life cycle cost of space missions modelled using Houbolt Questions. A case study of lunar cargo transport is presented in order to validate Houbolt questions as a method to modelling methodology

2. State of the Art

The state of the art of design space exploration, and early cost estimation of space mission architectures (MAs) are presented here.

Despite the existence of a theoretically infinite solution set, it is common that mission designers only consider a handful of alternative MAs in their trade-study during preliminary design of space missions Ross and Hastings write in *the Tradespace Exploration Paradigm* [5]. that "premature reduction of the [design-space] may take away valuable information from the designer, preventing realization of more robust and valuable systems."

Ross and Hastings' idea to avoid premature reduction of the design-space was Multi-Attribute Tradespace Exploration (MATE) [6]. In this work they describe a mixed integer and continuous method of discretising mission architectures.

The MATE method allows the designer to con-

sider many thousands of alternative mission architectures. However the design parameters are adapted to Earth-observation and communication constellation missions. In the literature, there is limited application of this method to exploration missions.

Recent work has been done at ISAE-Supaero to model mission architectures for space exploration. Beauregard et al in 2021 chose to iterate over trajectory choices and degrees of re-usability to create requirements for forty alternative mission architectures for a reusable lunar lander. This method was limited because it was performing computationally expensive trajectory optimisation on every candidate architecture, thus it could not consider a large number of them on the scale of MATE [7].

Ruede et al in 2023 proposed pattern languages as a new way to model the design-space of mission architectures [8]. A pattern language is a set of symbols and a set of grammatical rules which govern in which order symbols can appear. Much like words in a sentence, one can generate a combinatorial of every possible sequence of characters and examine it for interesting mission architectures. The pattern language proposed by Ruede is excellent at visualising the path and staging strategy of a mission architecture. However, it does not capture other important questions for the mission designer such as reusability considerations, subsystem choices, production methods and autonomy levels.

Ruede also mentions the concept of Houbolt questions and defines them as "key enabling questions" of the mission. A given example in the Apollo program is "to rendez-vous or not to rendez-vous in lunar orbit". Evidently this decision had a significant impact on the capabilities of the Apollo mission. The concept of the Houbolt question is not developed further by Ruede in [8].

Regarding cost estimation, there are established methods of estimating the cost of space missions in preliminary design phases. Currently, in space mission design, parametric cost models such as ECOM, NAFCOM and TruePlanning are used to estimate the costs of mission proposals by ESA during the preliminary design phase [9]. These models use cost estimation relationships (CERs) based on the complexity and mass of the subsystems and components to be produced.

Complexity can be estimated by the function of the subsystem, however, it is difficult to know the mass of systems in the early design stages. Arney and Wilhite in 2012 devised a method using graph theory to model the mission architecture space and to size the subsystems based on which branch of the graph they lie [10]. Statistically derived sizing rules and cost estimation relationships are used to size the subsystems over a limited range of inputs which are valid over the whole graph.

This state of the art has described two disciplines: design space exploration, and cost estimation. The contribution of this paper is adapting an implementation of both such that they can be linked into one methodology for designing low cost space missions.

3. Methodology

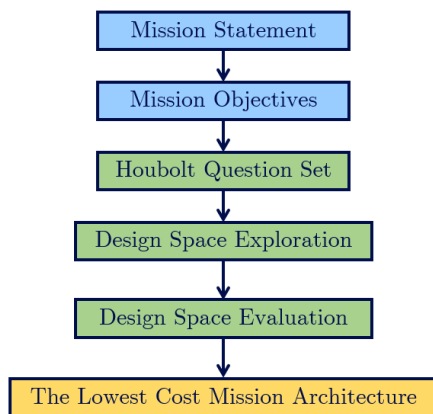


Figure 1: The Moneyball Methodology. Novel sections in green.

Figure 1 shows the adapted space mission design methodology which is the subject of this paper. Given a certain mission statement, and mission objectives, the mission designer should be able to model and explore the design space using Houbolt Questions. The models in the design space must contain enough information to enable evaluation in terms of cost.

Houbolt Questions are a proposed modelling framework for space mission architectures which uses a comprehensive list of design questions to describe space mission architectures. In this paper

Houbolt Questions						
Questions		Subject	Answers			
What	to send	mp	[kg]			
	to send back	mp returned	[kg]			
	Destination	Destination List (#32)	Lunar surface	Lunar orbit	Helocentric	...
How	to control it	Level of autonomy	E1	E2	E3	E4
	to launch it	Launch Strategy	Single Launch		Rideshare	
	to route it	Trajectory	Hohmann	Grav. assist	Low Thrust	Low Thrust + Grav. Assist
	to communicate with it	Comms link	Deep Space Network	Dedicated GS		Leased GS
	to dispose of it	EOL Strategy	Leave in Place	Passive 2nd Mission	Active 2nd mission	

Figure 2: The Houbolt Questions for a cargo delivery mission. Other sets exist for Exploration, and Sample return missions.

two types of Houbolt Questions are defined: *What Questions* and *How Questions*. The *What Questions* are set by the customer to define their initial needs. The *How Questions* are answered by the mission designer in order to produce a mission architecture model which has the lowest cost. A mission architecture in this paper is defined as the resulting combination of answers to the Houbolt questions.

The purpose of the Houbolt Questions is to describe mission architectures in a systematic way such that we can explore the design space of mission architectures. The subjects of the Houbolt questions are chosen for their impact on the life cycle cost of the mission.

$$LCC = C_d + C_p + C_o + C_{disp} \quad (1)$$

Where C_d is the design development and test engineering cost, C_p is the production cost, C_o is the operations cost and C_{disp} is the disposal cost. Each of these costs can be expressed as a function of parameters which are gleaned from Houbolt Questions as shown in figure 3.

		Costs			
		Development Cost	Production Cost	Operation Cost	Disposal Cost
Houbolt Questions	mp	$m_d = f(m_p, \Delta V, DRS)$ $\Delta V = f(Dest, Traj)$	$m_d = f(m_p, \Delta V, DRS)$ $\Delta V = f(Dest, Traj)$		
	mp returned	$m_d = f(m_p, \Delta V, DRS)$ $\Delta V = f(Dest, Traj)$	$m_d = f(m_p, \Delta V, DRS)$ $\Delta V = f(Dest, Traj)$		
	Destination List	$m_d = f(m_p, \Delta V, DRS)$ $\Delta V = f(Dest, Traj)$	$m_d = f(m_p, \Delta V, DRS)$ $\Delta V = f(Dest, Traj)$		
	Level of autonomy			$f(?)$	
	Launch Strategy		$imLEO = f(m_d, m_p, \Delta V)$		
	Trajectory	$m_d = f(m_p, \Delta V, DRS)$	$m_d = f(m_p, \Delta V, DRS)$		
	Comms link			$f(?)$	
	EOL Strategy				$f(?)$

Figure 3: The link parameters between information contained in the Houbolt questions and the life cycle cost of an MA.

To pass from the Houbolt Questions to the cost estimation, one needs some link parameters as shown in figure 3. Namely, the total dry mass launched into space for the mission m_d ; the initial mass to low earth orbit $imLEO$; The ΔV required to reach the destination.

To estimate these link parameters from the Houbolt Questions, one needs system to adapt sizing algorithms and design reference vehicles such as those described in [11]. These are spacecraft sizing rules which have been applied to lunar landers, interplanetary transfer vehicles, and orbital spacecraft.

For a given mission, the current rudimentary method for finding the lowest cost mission architecture proceeds as follows:

1. For the given mission, construct the design-space by listing all combinations of possible answers the the *How Questions*. Each combination is known as a "solution." and is saved to the solution database.
2. Calculate the "link parameters" between the Houbolt Questions and the cost elements shown in 3, using the sizing algorithms adapted from [11]. To calculate the dry mass of all the mission elements it will be necessary to choose reference spacecraft. Additionally it is also necessary to define how many spacecraft there should be through an optimal staging calculation.
3. Using the Advanced Mission Cost Model, calculate the cost elements from the link parameters

for every solution in the solution database.

4. Plot the total cost of all the solutions as points on a graph, find the index of the solution with the lowest life-cycle cost.
5. retrieve the solution from the database.

4. Case Study

This methodology was implemented on a use case. The use case mission statement is: *To land two tonnes of payload on the lunar surface.* This mission statement is used to set the *What Questions*:

$$m_p = 2000 \quad (2)$$

$$m_{p, returned} = 0 \quad (3)$$

$$Destination = (Lunar, Surface) \quad (4)$$

Before describing the answers to the how questions there are some constraints imposed by the study in order to feasibly demonstrate the capabilities of this methodology.

- The number of spacecraft carrying out the mission is set to 2: A Transfer Vehicle and a Lander.
- The Trajectory Design answer is set to Hohmann transfers only. Therefore we can say that the ΔV from Low Earth Orbit to the destination of $5800m/s$.
- This ΔV is fixed to each of the two spacecraft. $2000m/s$ for the lander and $3800m/s$ for the transfer vehicle.
- The level of Autonomy is not considered in this case due to the lack of a clear method to link the level of autonomy with the development and operations cost.

With these constraints, the design space is constructed from the combination of all the answers to the *How Questions* and all the possible choices of reference vehicles.

5. Results and Discussion

The design space with the current set of Houbolt questions small. For a given set of answers to the *What Questions*, the combinatorial of all the *How Questions* is nominally just 288. However when the reference vehicle choices for sizing the dry mass

are considered, the design-space grows to 1256 solutions. With such a small design space, the optimal solution is found by brute-force.

The initial masses to earth orbit, $imLEO$ for all solutions in the design space are plotted in figure 4.

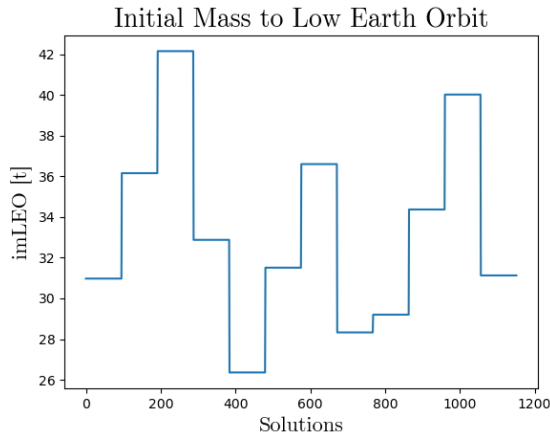


Figure 4: Initial Mass to Low Earth orbit

It is clear that many combinations are grouped together around the same $imLEO$. This is because some of the parameters such as the level of autonomy, and the trajectory are not affecting this variable. The main driver for the $imLEO$ is the combination of design reference vehicles which are being used. The lowest mass solutions (around solution no. 400) are all using LOX/LH2 powered Centaur-derived transfer vehicles and LOX/LH2 fuelled landers. These reference vehicles have the highest specific impulse and thus require the least propellant mass.

In figure 5m the system acquisition costs are plotted for all the solutions. The system acquisition cost ($C_d + C_p$) is the cost spent up to the operational life of the system. It is clear that there is some equivalence among the solutions because some of the Houbolt Question dominate in the cost estimation function. For example, the choice of design reference vehicles has much more influence than the communications architecture on the Acquisition cost.

This does not mean that the communications Houbolt Question can be ignored. The communications architecture can have a big impact on the operations cost, and in a commercial context where yearly cash flow is more important and since acquisition

costs can be amortised over several years, the operations cost of the system can become the dominant parameter in a cost optimisation.

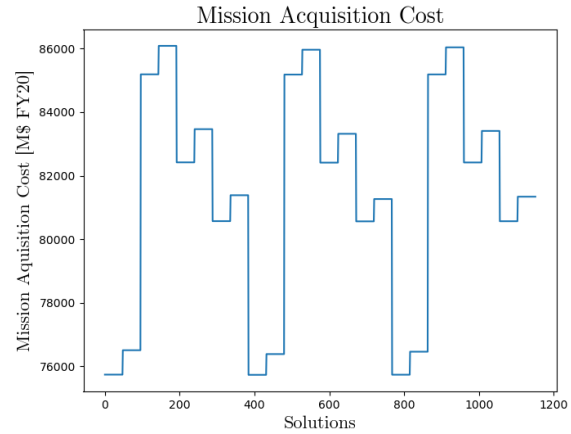


Figure 5: Mission acquisition costs

Figure 6 shows the same data as 5. Three main groupings of missions are apparent. The characteristics of these these groupings enables the identification of the most influential drivers of the acquisition cost.

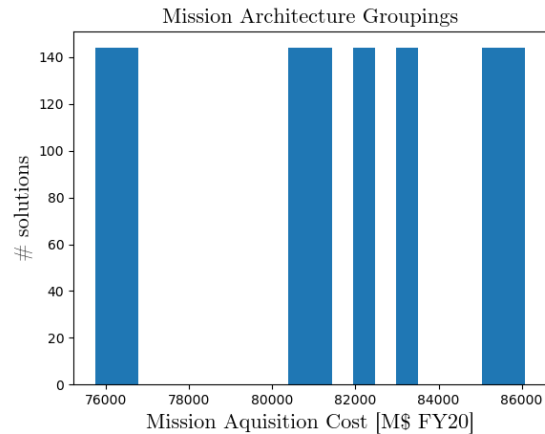


Figure 6: Groupings of mission architectures according to acquisition cost.

The first cheapest group are MAs using the Centaur-derived upper stage for their transfer vehicle. This first group includes landers of all types. the second middle grouping of three bars is the missions which use the Falcon 9 upper stage or the DCSS as their model transfer vehicles. The most expensive

group are those that use the Ariane 5 ECS as the transfer vehicle model.

The most expensive missions are using the transfer vehicles with the highest dry mass. This shows a shortcoming of the modelling method followed in this paper. The dry mass of the transfer vehicles dominates the other variables because the cost is estimated using the dry mass of the whole mission and the TV dry mass is the largest part of the total dry mass of the mission.

Two things can be done to improve this, the cost estimation of the dry mass of the transfer vehicles can be done separately to the lander since it is essentially an "of the shelf" spacecraft. Second, there should be a calculation to see if the mission architecture needs two spacecraft at all. In this case, the number of spacecraft was set to two, but a landing of cargo on the moon should in theory be possible with one spacecraft inserted directly to TLO by its launcher. Or perhaps it could be made more efficient by three spacecraft or more.

6. Conclusion

This paper has shown a first version of the Moneyball design space exploration method. The Houbolt Questions are shown to not be an absurd way to model space mission architectures such that their costs may be estimated early in the design.

The case study showed an implementation of the Moneyball method on a use case: deliver two tons of cargo to the lunar surface. The use case showed the capabilities of the methodology to consider many more alternative mission architectures than the state of the art. However, it also showed the method's limitations: The cost estimate of the mission is dominated by the choice of transfer vehicle model; and the method does not select the optimal number of spacecraft to complete the mission; and lastly, the methodology does not consider "depotting" solutions such as the lunar orbit rendezvous solution pushed by John Houbolt during the Apollo Program. The Jeep-problem offers an interesting avenue to explore depot placement in cis-lunar and interplanetary space.

Moreover the design-space constructed in the use case was quite small, around 1200 mission architectures. This was because of restrictions placed on the

question of trajectory. The ΔV parameter used to size the mission elements was set to a hohmann transfer value assuming impulsive maneuvers. It would expand the search space a great deal to consider different trajectories and different maneuvering strategies to reach the target.

A larger search space will make the brute-force method of finding the lowest cost mission architecture impossible. It would be interesting to see which optimisation methods will perform best at finding the lowest cost missions. Genetic algorithms are a candidate for this.

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